

Trustworthy Reasoning with Foundation Models: Robustness, Autonomy, and Interpretability

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Abstract. Large language models (LLMs) now solve many reasoning benchmarks, but in real use they can still fail when explanations are noisy, information is incomplete, or intermediate reasoning steps are hard to inspect. This thesis studies how to make LLM reasoning more *trustworthy* by improving reliability, decision-making autonomy, and interpretability. First, it introduces the NoRa benchmark and the CD-CoT method to measure and reduce performance drops caused by noisy chain-of-thought demonstrations. Second, it introduces AR-Bench to evaluate active reasoning, where models must ask useful questions and gather evidence before answering. Third, it proposes the Landscape of Thoughts (LoT) framework to visualize reasoning trajectories and support lightweight correctness verification. Together, these contributions provide practical benchmarks and methods for building trustworthy reasoning agents, and motivate a longer-term dual-system direction for scientific discovery.

Keywords: Trustworthy Machine Learning · Trustworthy Machine Reasoning · Foundation Models · Chain-of-Thought · Reasoning Trajectories

1 Research Questions and Significance

LLMs perform multi-step inference via chain-of-thought (CoT) and can act as reasoning agents; in real deployments, however, systems must remain reliable under imperfect inputs, act sensibly when information is missing, and remain diagnosable. This thesis investigates three research questions:

- **RQ1 (Robustness):** When CoT demonstrations contain *noisy rationales*—irrelevant or incorrect intermediate steps while the question and answer stay valid—how do models behave? Can we benchmark this setting and denoise with minimal supervision?
- **RQ2 (Autonomy):** How can we evaluate and improve *active* reasoning, where the model must ask questions and integrate evidence over multiple rounds, as opposed to *passive* reasoning with all information given upfront?
- **RQ3 (Interpretability):** How can we map reasoning trajectories to interpretable structures and use them for monitoring and verification?

Significance. Noisy rationales steer in-context learning and heuristics offer limited protection. Most benchmarks are passive; many real tasks need question-asking and evidence integration. Without trajectory-level measurement, grounded solutions are hard to distinguish from lucky guesses and verifiers for high-stakes applications are lacking.

2 Key Challenges and Related Work

Challenges. Noisy *rationales* (corrupted steps; questions and answers unchanged) are under-studied; inaccurate steps cause larger drops than irrelevant ones. Active reasoning needs benchmarks with incomplete problems and multi-round interaction; LLMs often ask poor questions or misuse evidence. Trajectory interpretability needs scalable, model-agnostic tools and verification without human process-level labels.

Related work. Chain-of-thought [3] and self-consistency [2] improve reasoning and search; in-context learning is sensitive to demonstration quality. Prior work on noisy inputs focuses on perturbed questions or valid/invalid demos rather than denoising corrupted rationales. Verification and process supervision improve reliability; ReAct [4] adds tool use and interaction. A unified benchmark for active reasoning and trajectory-level diagnostics has been lacking.

3 Main Contributions

3.1 Contribution I: NoRa & CD-CoT (NeurIPS 2024) [6]

We formalise *noisy rationales*: irrelevant or inaccurate intermediate steps in CoT demonstrations, with questions and answers unchanged. This setting is under-studied and harmful because corruption directly steers in-context learning. We construct NoRa, a benchmark of over 26,000 questions spanning mathematical (base arithmetic), symbolic (instruction mapping), and commonsense (family relations) tasks, with controlled noise ratios and both irrelevant and inaccurate thought types. Experiments show that even modest rationale corruption induces large accuracy drops (e.g., 3–33 percentage points for GPT-3.5), and that self-consistency and self-correction provide only limited improvement. We propose CD-CoT (Contrastive Denoising with noisy CoT), which uses a single clean rationale as supervision: for each noisy example, the model contrasts it with the clean reference to rephrase and denoise; we retain only rephrased examples whose answers remain correct, construct multiple prompting contexts, and aggregate final answers by majority vote. CD-CoT achieves about 18 percentage points average improvement over the base model and generalises to GSM8K, Blocksworld, and Hard Dyck without task-specific modifications.

3.2 Contribution II: AR-Bench (ICML 2025) [5]

We formalise *active* reasoning: the model receives an insufficient initial description and must interact over multiple rounds (e.g., with an answerer or oracle) to acquire evidence before producing a solution, in contrast to *passive* reasoning where all information is given upfront. We introduce AR-BENCH, comprising three task families: detective cases (interrogating suspects to identify the murderer), situation puzzles (yes/no questions to reveal a counterintuitive explanation), and guessing numbers (cracking a 4-digit code using symbolic feedback)—6,040 puzzles in total. We evaluate both *outcome* (final answer correctness) and *process* (quality of questions and evidence use across rounds). State-of-the-art LLMs (e.g., GPT-4o) achieve only 35% exact match on guessing numbers, whereas humans reach 80–100%; process scores improve rapidly in early

rounds then plateau; unreliable verifiers limit search-based methods (e.g., Tree-of-Thought); and SFT, DPO, Proactive CoT, and uncertainty-of-thought offer little or no benefit. AR-Bench motivates evaluation protocols and training objectives that explicitly model interaction, feedback, and partial observability.

3.3 Contribution III: LoT (ICLR 2026) [7]

We introduce LANDSCAPE OF THOUGHTS (LoT), a post-hoc, model-agnostic framework for analysing reasoning trajectories on multiple-choice tasks. The key idea is to characterise each intermediate state by a feature vector of perplexity-based distances to the answer choices, then to project these features into two dimensions (e.g., via t-SNE) to obtain a visual “landscape” of the trajectory. LoT pairs qualitative visualisations with quantitative metrics: *consistency*, *uncertainty*, and *perplexity*. We show that correct trajectories converge to the right answer later and more consistently, whereas incorrect ones often converge more quickly to wrong answers—a finding we verify statistically. A lightweight verifier (e.g., random forest) trained on these features predicts correctness without human process-level labels and consistently improves voting accuracy and test-time scaling across model sizes (1B–70B), reasoning methods (CoT, least-to-most, ToT, MCTS), and datasets (AQuA, MMLU, StrategyQA, CommonsenseQA). The remaining work is to finalise the thesis narrative and to proceed to submission and defence. Beyond the thesis, the research plan points toward a dual-system framework (reasoning [8] and measurement [1]) for scientific discovery.

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